Comparative Analysis of E. Coli and Enterococci Concentrations of two San Francisco Recreational Lakes

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Air & Water Group Projects, November 2025

Abstract

This study compared *E. coli*, enterococci concentrations, and turbidity levels in two San Francisco recreational lakes located in neighborhoods with differing median incomes. The goal was to assess whether lower-income communities have access to clean recreational water. Results showed higher turbidity in Blue Heron Lake, while McNab Lake displayed higher E. coli and enterococci levels. Both lakes exceeded recreational water quality standards for these indicators, highlighting the need for regular monitoring to ensure safe public access.

Hypothesis

A lake in a lower income neighborhood, McNab Lake, will have higher fecal coliform & E. coli levels, and turbidity compared to Blue Heron lake in Golden Gate Park, a higher income area.

Background/Introduction

Environmental factors such as water quality vary across the United States, and poorer water quality often correlates with community demographics (NAACP, 2024). This raises concerns about environmental justice, defined as the principle that all people have the right to safe, healthy environments with equal environmental protections and meaningful involvement. Prior research shows that lower-income communities frequently have reduced access to clean recreational waters. Wang et al. found that economic disparities were significantly associated with *E. coli* contamination in recreational waterbodies in Texas (Wang, Runzi, et al., 2024)

This study examines whether concentrations of *E. coli*, enterococci, and turbidity levels differ between two recreational lakes in San Francisco. Blue Heron Lake provides rowboat and pedal boat rentals (Blue Heron Boathouse, 2024), while McNab Lake allows canoeing (Wilderness Inquiry, 2025). Although neither lake permits swimming, both are used for recreational activities. The lakes differ in the median income of surrounding census blocks: those near McNab Lake average about $86,801, while those near Blue Heron Lake average approximately $122,461.25 (SF.gov, 2020). This study compares water quality in these two neighborhoods to assess disparities in access to clean recreational waters.

Methods & QA/QC

- Census data were used to determine which lakes were near lower and higher income neighborhoods, data from the eight closest census blocks to each lake were recorded and then the average of the 8 incomes were recorded.
- Five evenly distributed sites were sampled in each lake, with two samples collected per site. An additional two duplicate samples were taken from Site 1 in each lake, resulting in a total of 22 samples.
- Turbidity was measured at each site using a Hach 2100Q turbidity meter (2100, Hach Company 2024).
- Back at the lab the samples were tested for *E. Coli* and enterococci using the EPA approved colilert-18 and enterolert test procedures (9223, Hach Company 2024).
- Turbidity standards and field blank were measured on site for quality assurance.
- Results were recorded and tested for statistical significance using python 3.13.7.

Results

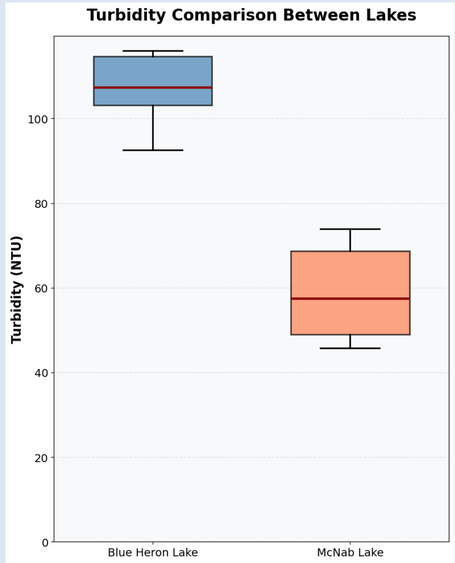

Figure 1: Box Plot of Turbidity Comparison

Blue Heron Lake had statistically higher turbidity than McNab Lake. (**t = 7.227, p = 0.0000**).

Average Turbidity Data Values (NTU)

	Blue Heron Lake	McNab Lake
Site 1	114.6	71.6
Site 2	107.3	73.9
Site 3	455.0	54.9
Site 4	103.1	47.0
Site 5	92.6	45.8
Site 1 Dup	116.0	60.0

Table 1: Turbidity data points, Site three is a **Statistical outlier by IQR method (>Q3 + 1.5 × IQR), therefore not included.**

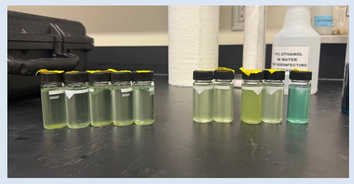

Figure 4: Turbidity vials for blue heron lake (right) and McNab lake (left).

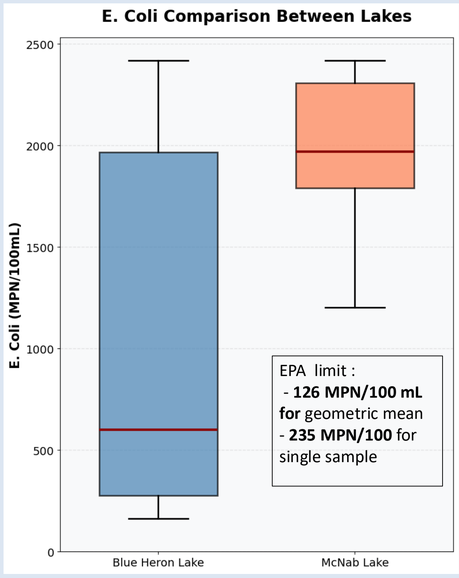

Figure 2: Box Plot of E. Coli Comparison

McNab Lake had higher E. coli levels than Blue Heron Lake, but this difference was not statistically significant **(t = -1.874, p = 0.0905)**.

E. Coli Data Values (MPN/100 mL)

	Blue Heron Lake	McNab Lake
Site 1	2419.6	1968.3
Site 2	613.1	2419.6
Site 3	172.3	1732.9
Site 4	161.6	1203.3
Site 5	589.4	1968.3
Site 1 Dup	2419.6	2419.6

Table 2: E. Coli data points.

Figure 5: Enterococci IDEXX quanti tray under fluorescent for McNab site 5

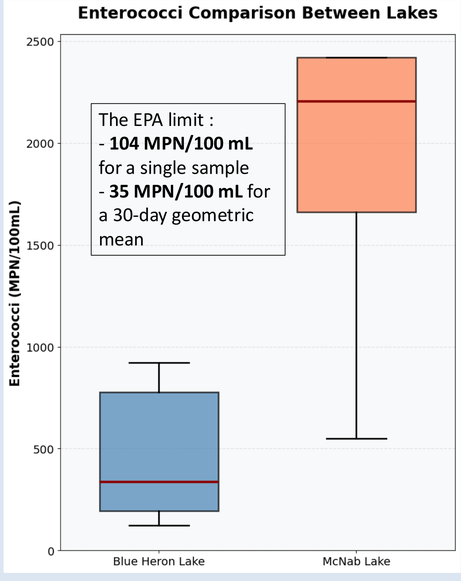

Figure 3: Box Plot of Enterococci Comparison

McNab Lake showed significantly higher enterococci levels than Blue Heron Lake **(t = -4.224, p = 0.0018).**

Enterococci Data Values (MPN/100 mL)

	Blue Heron Lake	McNab Lake
Site 1	920.8	1986.3
Site 2	328.2	>2419.6
Site 3	148.3	1553.1
Site 4	122.2	550.4
Site 5	344.8	>2419.6
Site 1 Dup	920.8	2419.6

Table 3: Enterococci data points.

Discussion & Conclusions

This study partially supported the hypothesis that McNab Lake, surrounded by lower-income neighborhoods, would have higher bacterial contamination and turbidity compared to Blue Heron Lake in Golden Gate Park. McNab Lake showed significantly higher enterococci levels and elevated E. coli concentrations, though the latter was not statistically significant. Contrary to predictions, Blue Heron Lake exhibited significantly higher turbidity.

Most critically, both lakes exceeded EPA recreational water quality standards for *E. coli* and enterococci, raising public health concerns (EPA, 2012). Sampling occurred following a rainstorm, which likely elevated bacterial levels due to urban runoff carrying fecal contamination from surrounding areas. While this may not represent typical conditions, the extreme values observed, with some samples reaching the upper detection limit of 2419.6 MPN/100mL, highlight a critical gap in water quality monitoring.

These findings underscore the need for regular, systematic testing of recreational water bodies in San Francisco parks. The exceedance of the EPA standards represents an environmental justice issue, as communities of all income levels deserve equal access to clean recreational waters. Future research should include longitudinal sampling across different weather conditions and seasons to better characterize baseline water quality and identify specific contamination sources requiring intervention. The results were consistent with those of Wang et al., (2024) although fewer sites were analyzed. To fully assess water quality equity in San Francisco, a larger-scale study comparable in scope to Wang et al.'s work should be conducted to observe correlations between different income and access to clean recreational waters .

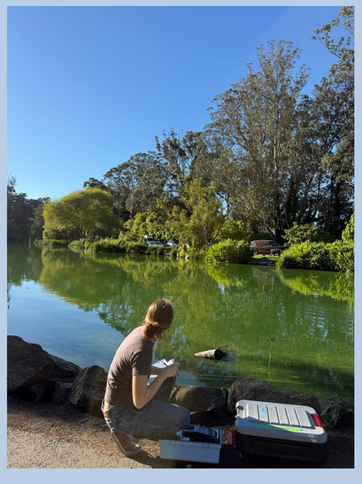

Figure 6: Turbidity testing at Blue Heron Lake on October 29th 2025

Figure 7: Turbidity testing at McNab Lake on October 29th 2025

References

United States Environmental Protection Agency. *Recreational Water Quality Criteria*. EPA, 2012, www.epa.gov/sites/default/files/2015-10/documents/rwqc2012.pdf.

City and County of San Francisco. *Demographics and Health Maps*. SF.gov, www.sf.gov/demographics-and-health-maps. Accessed 12 Nov. 2025.

Wang, Runzi, et al. "Environmental Justice of Texas Recreational Water Quality – The Disproportionate E. coli Levels and Trends." *Journal of Environmental Management*, vol. 371, 2024. *ScienceDirect*, www.sciencedirect.com/science/article/pii/S0301479724029554.

Blue Heron Boathouse, 9 Sept. 2024, blueheronboathouse.com/.

Environmental & Climate Justice Issue Brief: Clean Water." *NAACP*, 15 Feb. 2024, https://naacp.org/resources/environmental-climate-justice-issue-brief-clean-water

"McNab Lake." *Wilderness Inquiry*, 11 Mar. 2025, www.wildernessinquiry.org/itinerary/lake-mcnab/.